NovaTech Systems — NovaID Technical Documentation

NovaTech Systems • Internal Knowledge Base Document

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Purpose and scope

Technical reference for NovaID authentication, authorization, roles, permissions, OAuth 2.0 concepts, service identity, and troubleshooting.

All names, systems, processes, dates, limits, and examples in this document are fictional and exist only for the NovaTech Systems RAG knowledge-base exercise.

1. Purpose

NovaID is NovaTech's internal identity and access-management service used by NovaTech products and internal systems. It centralizes authentication, authorization, role assignment, and service identity.

2. Architecture Overview

A typical request is authenticated through NovaID, receives an access token representing the authorized identity and scope, and is evaluated by the target service before protected operations are permitted.

Component	Responsibility
Authentication	Verifies the user or service identity through approved identity flows.
Authorization	Determines whether the identity has the required role/permission.
Access token	Short-lived credential presented to protected APIs.
Refresh token	Longer-lived credential used to obtain a new access token where the approved OAuth flow permits it.
Role	Named bundle of permissions representing a common access profile.

3. OAuth 2.0

NovaID supports OAuth 2.0 patterns for delegated authorization and token-based API access. Client applications must use approved flows and must not expose client secrets or tokens to unauthorized parties.

4. Example Roles

Role	Typical purpose
ADMIN	Administrative management and privileged configuration
ENGINEER	Engineering operations and approved development access
SUPPORT_AGENT	Customer/service support operations
ANALYST	Analytics and reporting access
VIEWER	Read-only access where supported

5. Permissions

Permissions are the granular authorization units used by services. A role may map to multiple permissions. Applications should check the required permission at the operation boundary rather than assuming that authentication alone grants access.

6. Service-to-Service Authentication

- Services authenticate using approved service identities rather than employee credentials.
- Service credentials must be stored in the approved secrets-management system.
- Services should request only the permissions required for their function.
- Rotate or revoke compromised service credentials according to NT-SEC-001.

7. Common Errors

Error	Likely meaning	Initial troubleshooting
401 Unauthorized	Missing, invalid, or expired authentication	Check token presence, issuer, expiry, and authentication flow.
403 Forbidden	Authenticated identity lacks required permission	Check role/permission mapping and resource authorization.
invalid_token	Token cannot be accepted by the resource	Check expiry, signature validation, issuer, audience, and token format.
invalid_scope	Requested scope is not permitted	Check client configuration and approved API scopes.

8. Troubleshooting Procedure

1. Identify whether the failure is authentication (401) or authorization (403).
2. Confirm the client is using the intended NovaID environment.
3. Inspect non-secret token claims such as issuer, audience, expiry, and scopes.
4. Verify that the calling identity has the required permission.
5. Check service logs and correlation identifiers.
6. Escalate persistent identity-service failures to the Identity team.

9. Security Notes

Never place real credentials, tokens, private keys, or production secrets in documentation, tickets, source code, or test examples. This knowledge-base document intentionally contains no real credentials.

10. Related Documents

- NT-SEC-001 — Security Policy

- NT-ENG-001 — Engineering Handbook
- NT-IT-001 — IT Support Guide
- NT-HR-005 — Employee Onboarding Guide